



AL FLASHER 12 V ENDURANCE TEST REPORT

DATE:16.04.2026

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Part Name	AL FLASHER 24 V	Application	COMMERCIAL APPLICATION
Customer Part No.	FGG00700	Customer	ASHOK LEYLAND
UCAL Part No.	RDEPL/AL/16	Report No	R&D/AL/24VF/VAL/26/002

INTRODUCTION

UCAL is currently supplying 24V AL Flashers to Ashok Leyland for commercial applications. This validation plan has been prepared to define the test requirements and validation activities for the current production parts of the respective product. The testing will be carried out as per the approved validation plan to verify product performance, functional reliability, and compliance with specified requirements under defined test conditions

PURPOSE:

To evaluate the 24 V flashers assembly after **Endurance Test**.

TEST SPECIFICATION:

Test standard as per **ISO 16750-2:2023**. DUT should operate within specified functional parameters.

ACCEPTANCE CRITERIA:

Class A: All function of the flasher performs as designed during and after the test.

RESULT:

SAMPLE NO	PERFORMANCE	APPEARANCE	JUDGEMENT
S:03	O	O	O

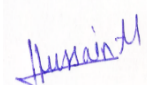
O: OK

X: NG

Δ: To be confirmed

CONCLUSION:

The performance of the Flasher assembly is acceptable after test. The samples are met the requirements.

 Mr. Rahul Bongarde	 Mr. Hussain Merchant
Prepared By	Approved By



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PERFROMACE TEST SPECIFICATIONS:

SNO	PARAMETERS	SPECIFICATIONS
1	Operating voltage	24 V~32 V
2	Test voltage	24 V
3	Flash rate(Turn signal/Hazard warning)	60~120 Flashes/Minute
4	One lamp failure flashing rate	>140 Flashes/Min(or)1.8 times of flash rate whichever is greater
5	Duty Cycle	40% ~60%

TEST PROCEDURE:

Standard test circuit with a test voltage of 24 V shall be applied for flasher with operation.

Temp: Ambient.

Hazard: Continuous operation with rated load for 36hours

Direction Indicator: Continuous operation with rated load for 100hours.

Temp: $80 \pm 2 \text{deg}^\circ\text{C}$

Hazard: Continuous operation with rated load for 5hours

Direction Indicator: Continuous operation with rated load for 30hours.

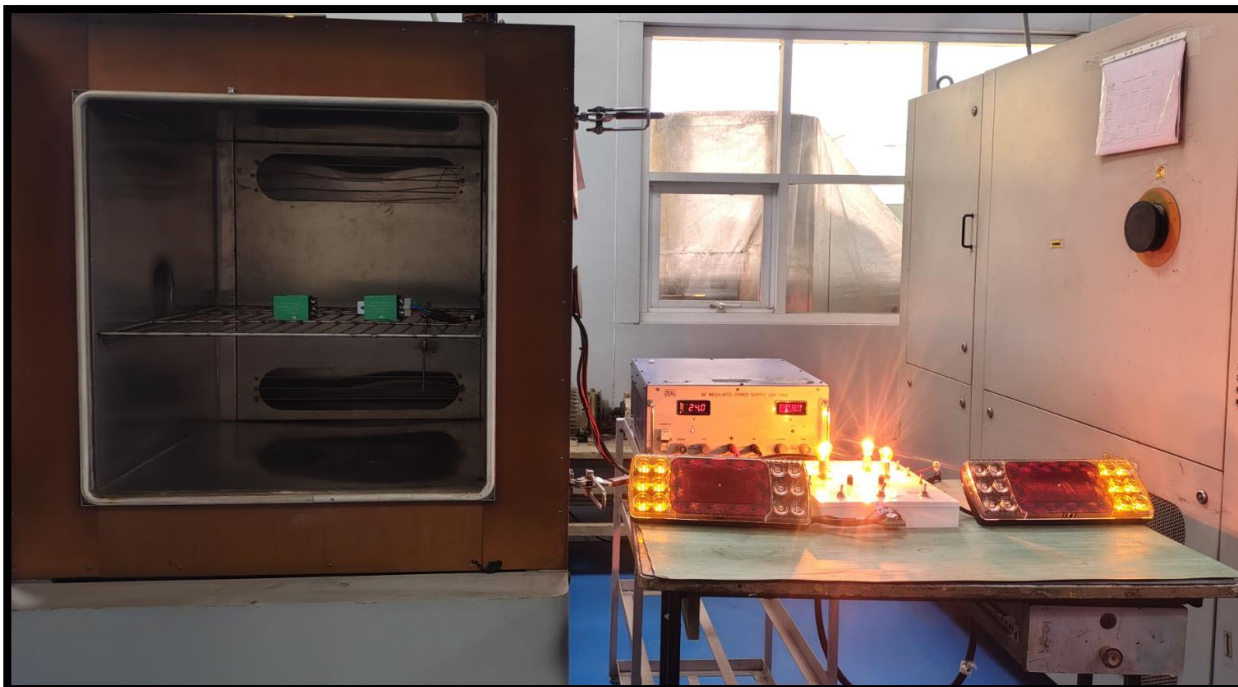
Temp: $-10 \pm 2 \text{deg}^\circ\text{C}$

Hazard: Continuous operation with rated load for 5hours

Direction Indicator: Continuous operation with rated load for 30hours.

Total Duration:206hrs.

TEST SETUP:



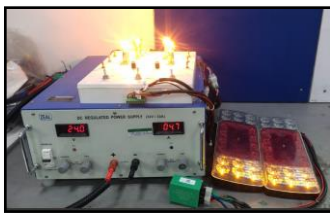


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TURN SIGNALS & HAZARD SIGNALS:

LEFT SIGNAL	RIGHT SIGNAL	HAZARD SIGNAL
		

TEST OBSERVATION:

Description	Load	Observations(at 12 V)	Remark
	LED& CLUSTER	Sample No:01	
Turn Signal RH	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	85	OK
Turn Signal LH	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	85	OK
Hazard Mode	2x (2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal)	85	OK
Fault condition Any one 21W Lamp failure	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	171	OK
Duty cycles for RH&LH	2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal	49%	OK
Duty cycles for Hazard Mode	2x (2 x 21 W Lamp + 1x 3.5 W LED+ 2 W Cluster Signal)	49%	OK

NOTE: The observed flashes are within the specifications & the samples met the performance requirem

TESTED SAMPLES PHOTOS:

BEFORE TEST SAMPLE – S:03	AFTER TEST SAMPLE – S:03
	

OBSERVATION:

There is no significant change in the electrical performance of the flasher samples. The functionality remains satisfactory. The observed flashes are within the specified requirements & the samples met the performance requirements. All function of the Flasher performs as designed before & after the test.